

PMorph: A Task-Specialized Robot Manipulator Morphology Optimizer

Steve Gillet

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Problem



Humans



The End

Problem

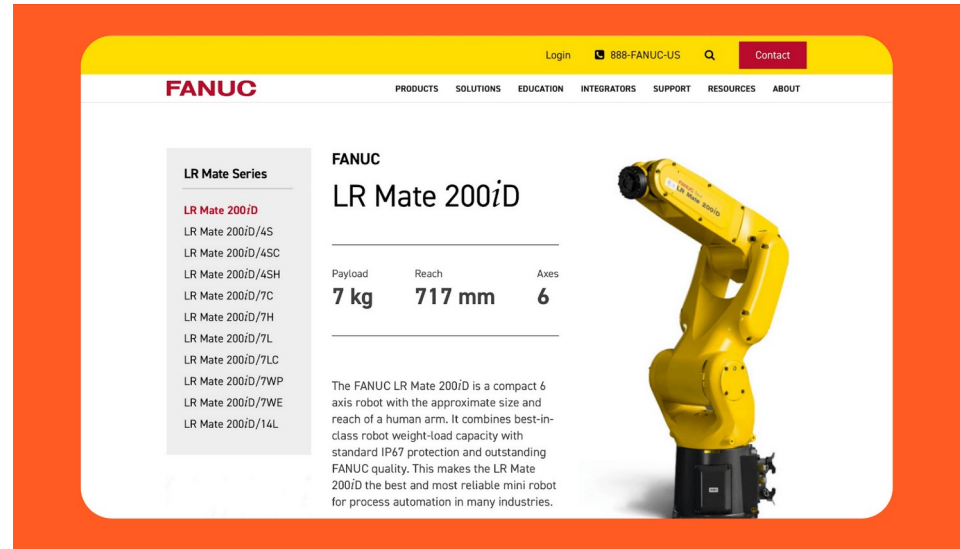
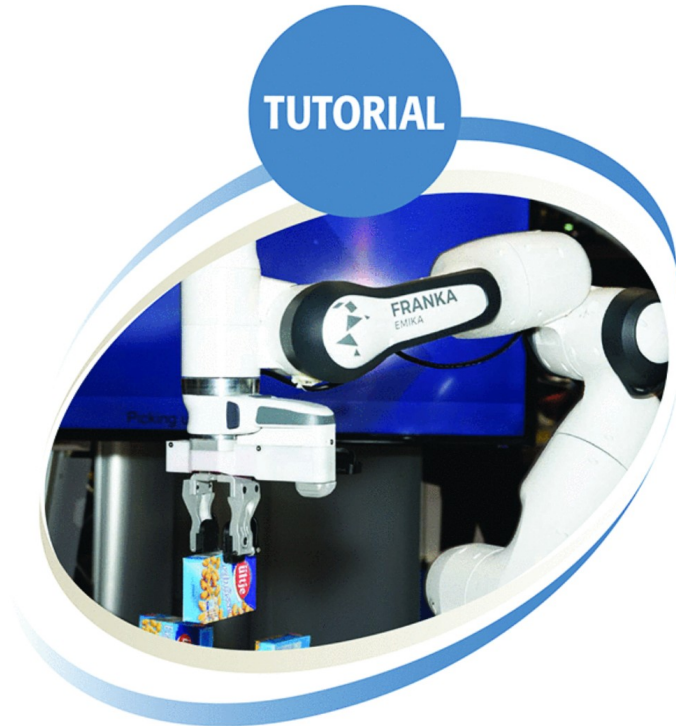


- Human workers specialize
- Robots don't

[1] Madieto et al., Prod. Oper. Manag., 2020.

[2] Patel, IOSR J. Eng., 2024

Generalized Robotic Manipulators



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LR Mate Series

- LR Mate 200iD
- LR Mate 200iD/4S
- LR Mate 200iD/4SC
- LR Mate 200iD/4SH
- LR Mate 200iD/7C
- LR Mate 200iD/7H
- LR Mate 200iD/7L
- LR Mate 200iD/7LC
- LR Mate 200iD/7WP
- LR Mate 200iD/7WE
- LR Mate 200iD/14L

FANUC LR Mate 200iD

Payload	Reach	Axes
7 kg	717 mm	6

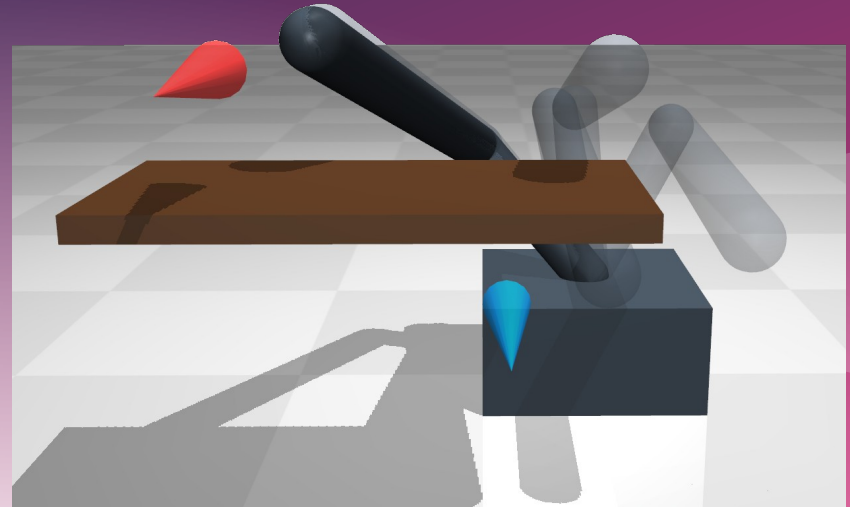
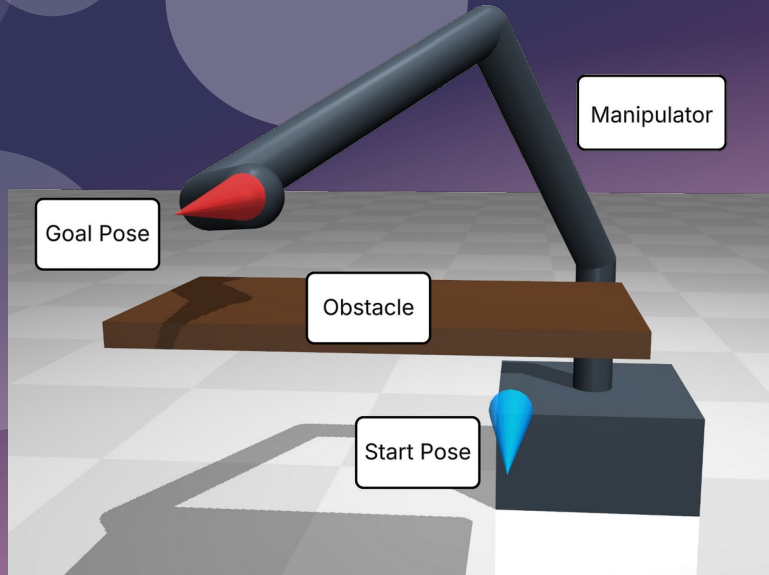
The FANUC LR Mate 200iD is a compact 6 axis robot with the approximate size and reach of a human arm. It combines best-in-class robot weight-load capacity with standard IP67 protection and outstanding FANUC quality. This makes the LR Mate 200iD the best and most reliable mini robot for process automation in many industries.

[3] Haddadin et al., IEEE Robot. Autom. Mag., 2022.

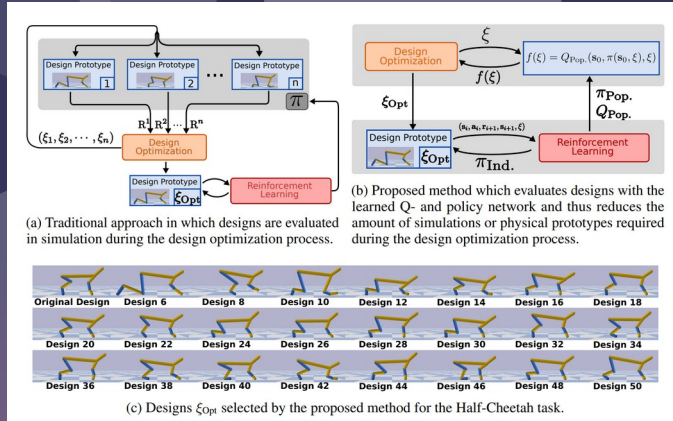
[4] Standard Bots, 2025.

[5] He et al., Int. J. Adv. Robot. Syst., 2019.

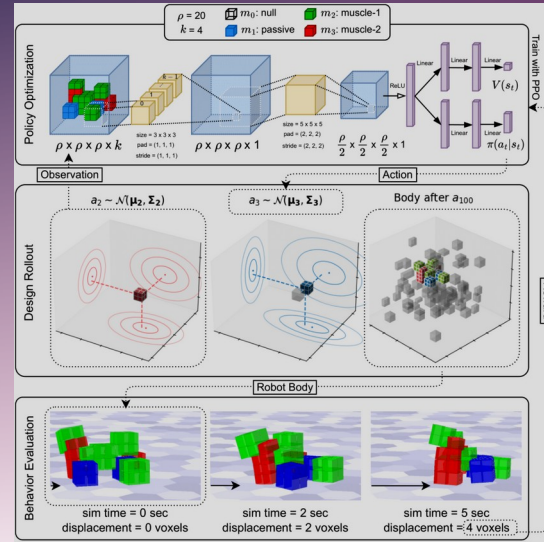
Solution



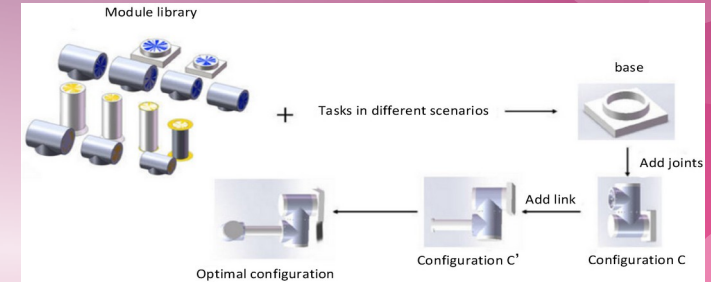
Related Work



[6] Luck et al., CoRL, 2019.



[7] Bhatia et al., arXiv:2310.05670, 2023.

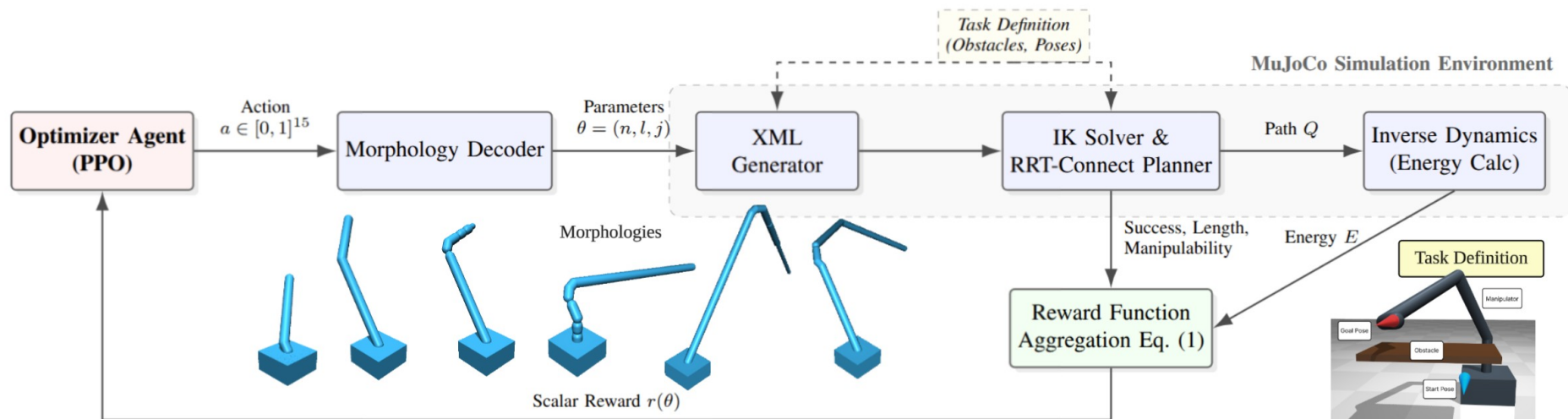


[8] Ding et al., Symmetry, 2024.

Research Questions

- 1) Can PPO be used to generate more efficient manipulator morphologies?
- 2) Can morphologies be generated for tasks that generalized manipulators cannot do or cannot do well?
- 3) Can more simple morphologies (less DOF) be generated that can perform the same tasks with similar levels of accuracy?

PMorph

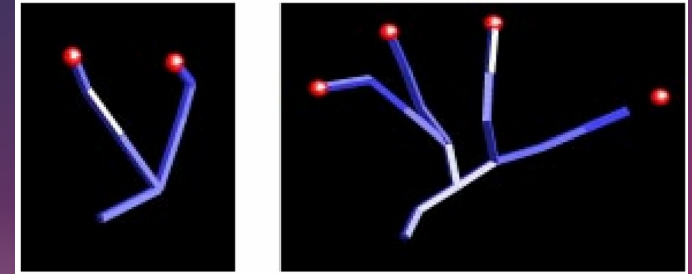
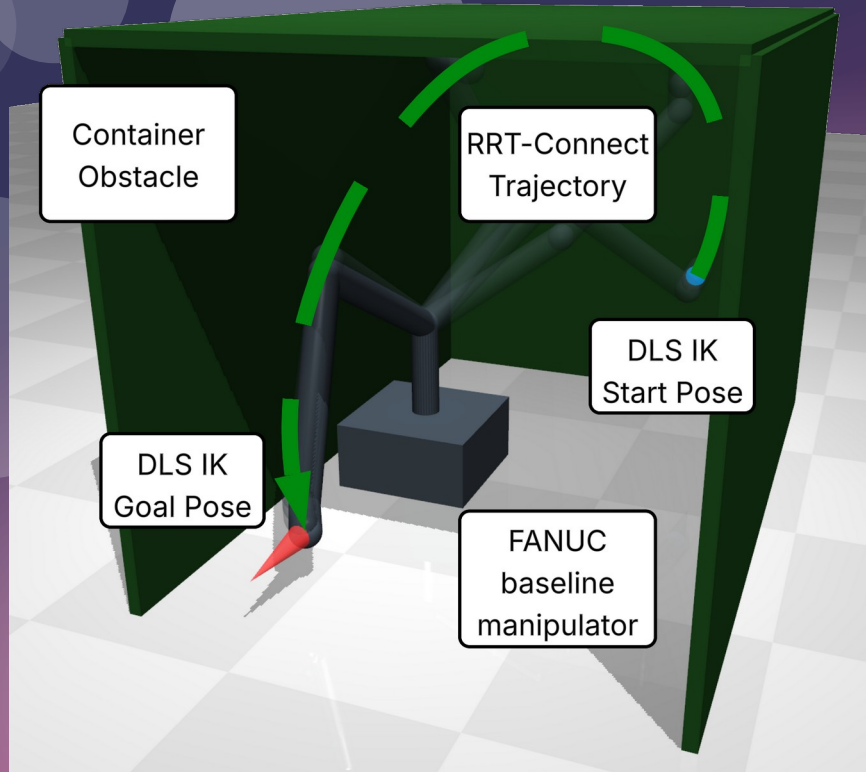


Methodology

Algorithm 1 Morphology Decoding from PPO Action

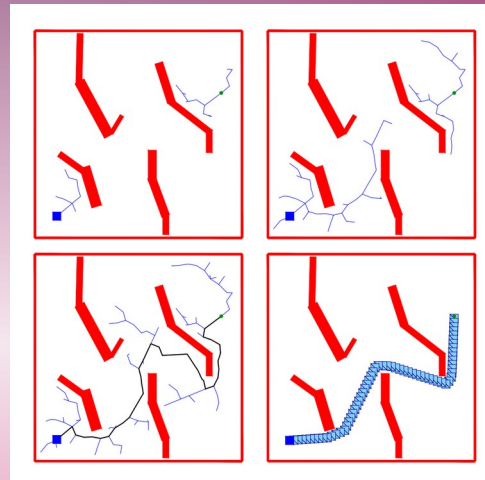
- 1: **Input:** action $a \in [0, 1]^{15}$
 - 2: $n \leftarrow \text{round}(a_0 \cdot (n_{max} - n_{min}) + n_{min})$ ▷ links: 2–7
 - 3: $l \leftarrow a_{1:8} \cdot (l_{max} - l_{min}) + l_{min}$ ▷ lengths: 0.05–1.2m
 - 4: $j \leftarrow \text{round}(a_{8:15} \cdot 3)$ ▷ types: 0–3 (hinge X/Y/Z, slide Z)
 - 5: **Return:** $\theta = (n, l_{:n}, j_{:n})$
-

Simulation



$$\|J\Delta\theta - \vec{e}\|^2 + \lambda^2\|\Delta\theta\|^2$$

[9] S. R. Buss, UCSD Tech. Report, 2004.



[10] Kuffner and LaValle, ICRA, 2000.

Reward Calculation

Balance Scale

Capabilities (Weighted Positively)



Task Success
(Valid path mapped
via RRT-Connect)



Manipulability Index (μ)
at start and goal

[WEIGHT_POS: 2.4]

Scalar Reward $r(\theta)$

Efficiency Penalties (Weighted Negatively)



Energy Cost
(E via inverse dynamics)



Link Count
(n, heavily penalizing >2 links)



Kinematic Error
(e at start and goal)



Path Length (p)

[WEIGHT_NEG: -5.1]

$$r(\theta) = \frac{1}{N} \sum_{i=1}^N \left[w_0 \cdot \text{success}_i - w_1 \cdot p_i - w_2 \cdot (e_{\text{start},i} + e_{\text{goal},i}) + w_3 \cdot (\mu_{\text{start},i} + \mu_{\text{goal},i}) - w_4 \cdot (n - 2) - w_5 \cdot E_i \right]$$

PPO Training

Algorithm 2 PMorph Training

```
1: Initialize PPO policy  $\pi_\phi$ 
2: for each episode  $t = 1$  to  $1M$  do
3:   Sample  $\theta \sim \pi_\phi(\cdot|\phi)$ 
4:    $r \leftarrow \text{eval}(\theta)$  ▷ Section 3 pipeline
5:   Store  $(\theta, r)$  in buffer
6:   Update  $\phi$  via the clipped surrogate loss [43]:

$$\mathcal{L}(\phi) = \hat{\mathbb{E}}_t \left[ \min(r_t(\phi) \hat{A}_t, \right.$$


$$\quad \left. \text{clip}(r_t(\phi), 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right]$$


$$+ S[\pi_\phi](s_t) - \beta \cdot \mathbb{E}_t[(\hat{V}_\phi(s_t) - V_t^{\text{targ}})^2]$$

▷  $\epsilon = 0.2$ , GAE advantages [42]
7: end for
```

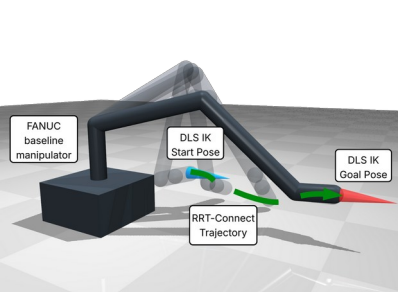
[43] Schulman et al.,
arXiv:1707.06347,
2017.

[42] Schulman et al.,
arXiv:1506.02438, 2015.

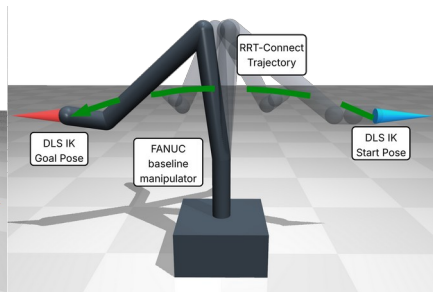
Hyperparameters: $\epsilon = 0.2$ (clip ratio) and $\beta = 0.5$ (value loss weight)

Evaluation

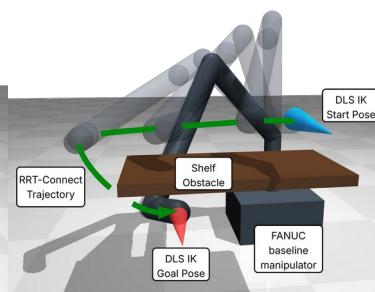
Outreach



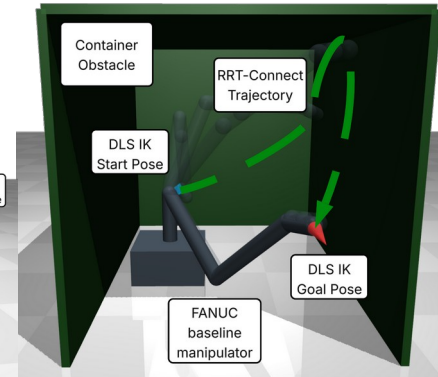
Side-to-side



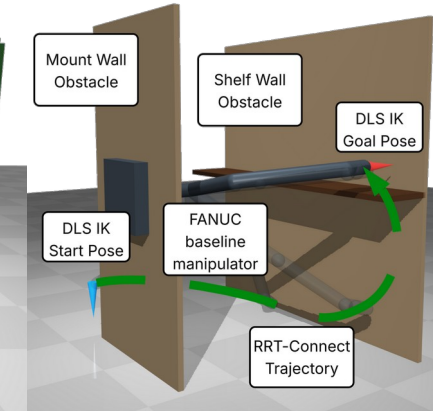
Shelf



Container



Wall mount



SAC

[13] Haarnoja et al., ICML, 2018.

CMA-ES

[14] Hansen and Ostermeier, Evol. Comput., 2001.

Results

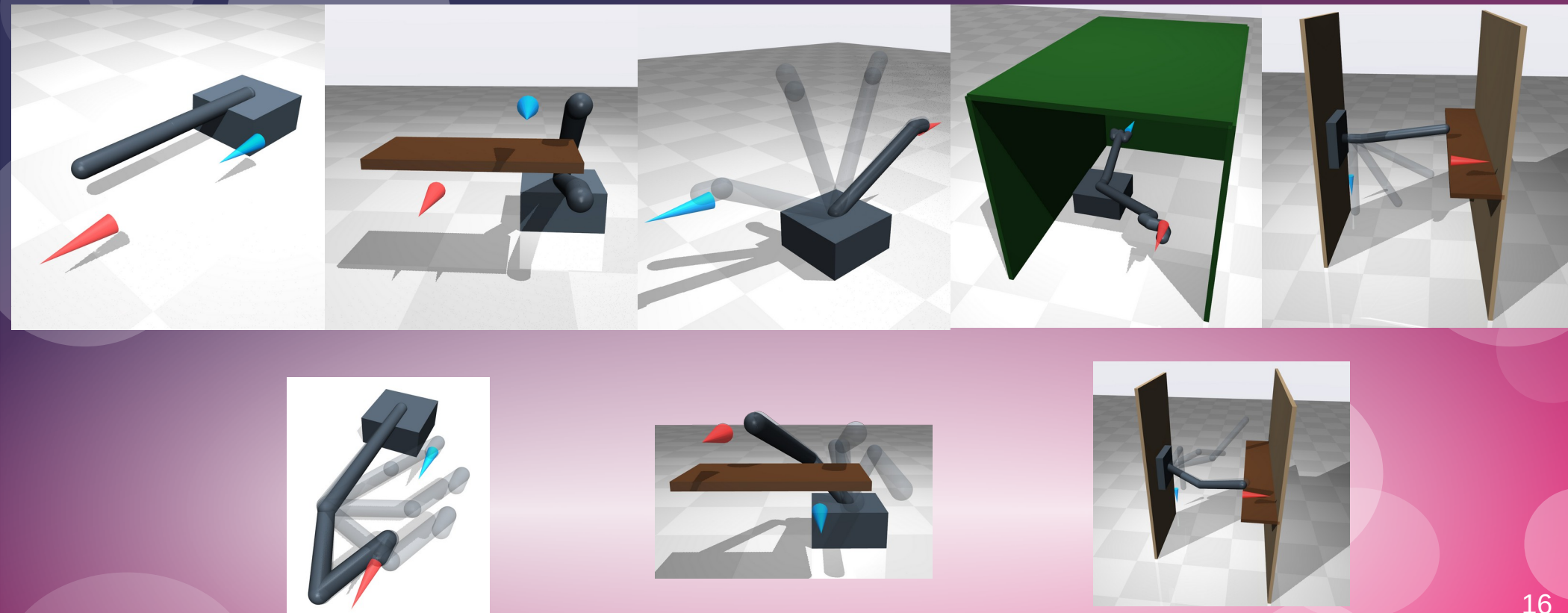
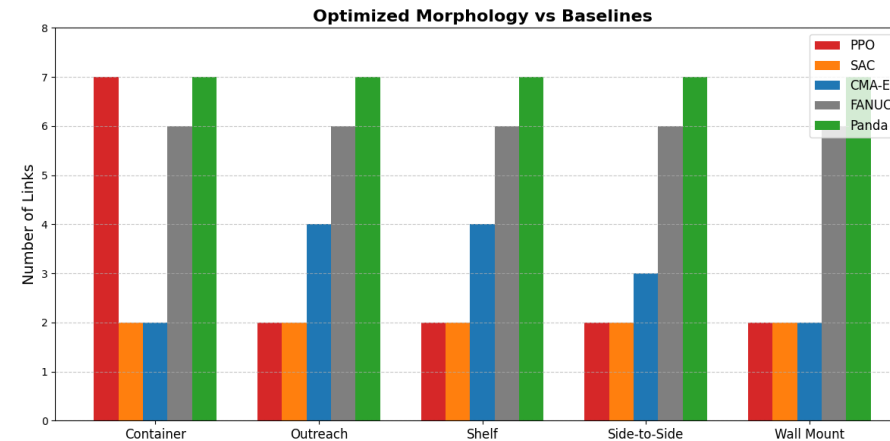
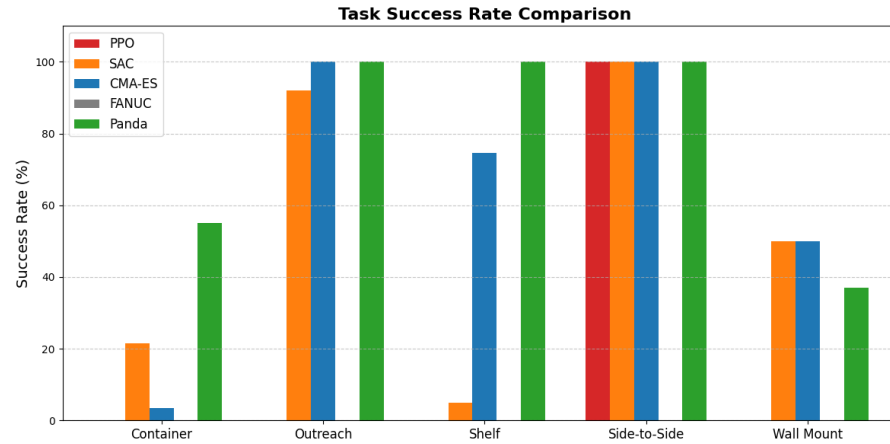


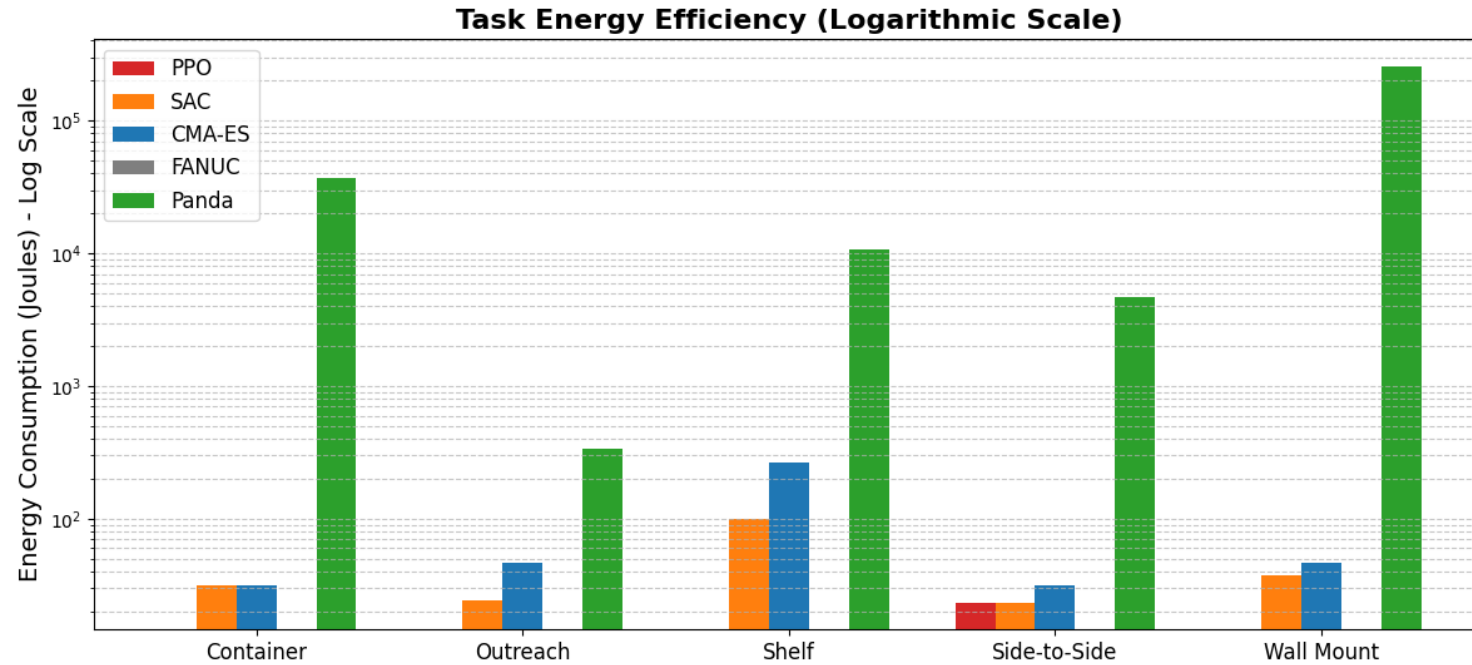
Table 6.1: Performance Metrics Across All Tasks and Morphologies (Averaged over 100 runs)

Task	Model	Links (n)	Success (%)	Pos Err (m)	Rot Err (rad)	Path Len (m)	Energy (J)	Manip. (μ)
Container	PPO	7	0.0	0.039	0.279	∞	∞	0.000
	SAC	2	21.5	0.086	0.433	1.600	31.87	0.000
	CMA-ES	2	3.5	0.085	0.420	1.541	31.40	0.000
	FANUC	6	0.0	0.032	0.474	∞	∞	0.185
	Panda	7	55.0	0.001	0.312	2.055	36898.65	0.224
Outreach	PPO	2	0.0	0.220	0.031	∞	∞	0.000
	SAC	2	92.0	0.050	1.380	0.521	24.22	0.000
	CMA-ES	4	100.0	0.011	0.042	0.554	46.29	0.000
	FANUC	6	0.0	0.003	0.406	∞	∞	0.144
	Panda	7	100.0	0.001	0.360	2.072	337.13	0.220
Shelf	PPO	2	0.0	0.230	0.113	∞	∞	0.000
	SAC	2	5.0	0.113	0.586	0.840	100.36	0.000
	CMA-ES	4	74.5	0.048	0.751	4.025	265.16	0.000
	FANUC	6	0.0	0.064	0.659	∞	∞	0.171
	Panda	7	100.0	0.002	0.332	2.828	10718.82	0.198
Side-to-Side	PPO	2	100.0	0.029	0.026	0.987	23.50	0.000
	SAC	2	100.0	0.028	0.026	0.989	23.49	0.000
	CMA-ES	3	100.0	0.010	0.031	1.005	31.31	0.000
	FANUC	6	0.0	0.002	0.375	∞	∞	0.171
	Panda	7	100.0	0.002	0.413	2.331	4669.10	0.157
Wall Mount	PPO	2	0.0	0.384	0.090	∞	∞	0.000
	SAC	2	50.0	0.077	1.096	1.827	37.29	0.000
	CMA-ES	2	50.0	0.077	1.098	1.870	46.40	0.000
	FANUC	6	0.0	0.070	1.594	∞	∞	0.116
	Panda	7	37.0	0.024	1.767	3.056	257456.71	0.189

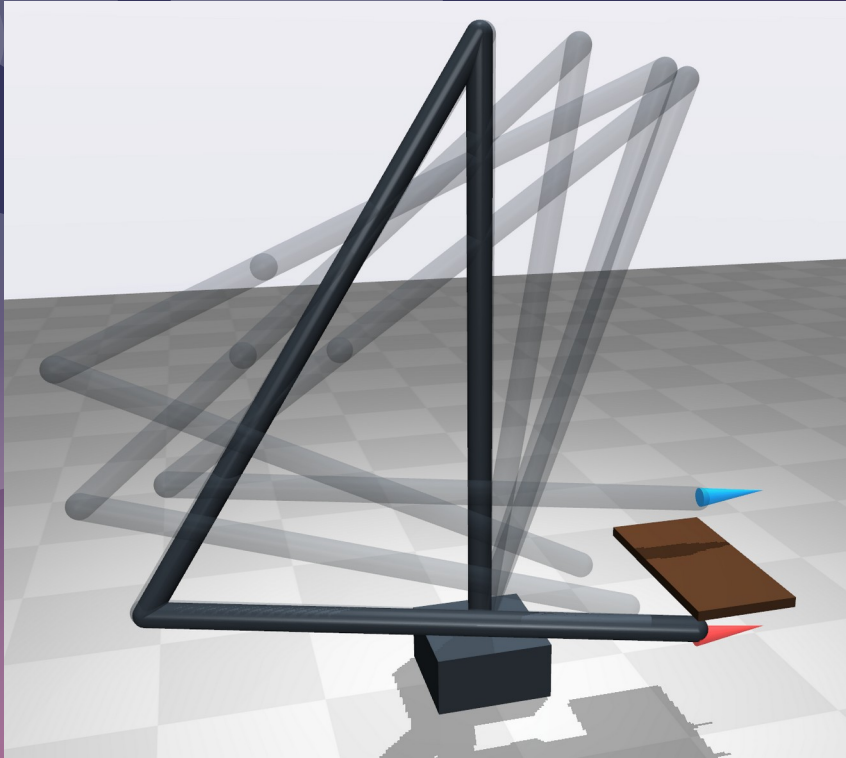
Specialization & Simplicity



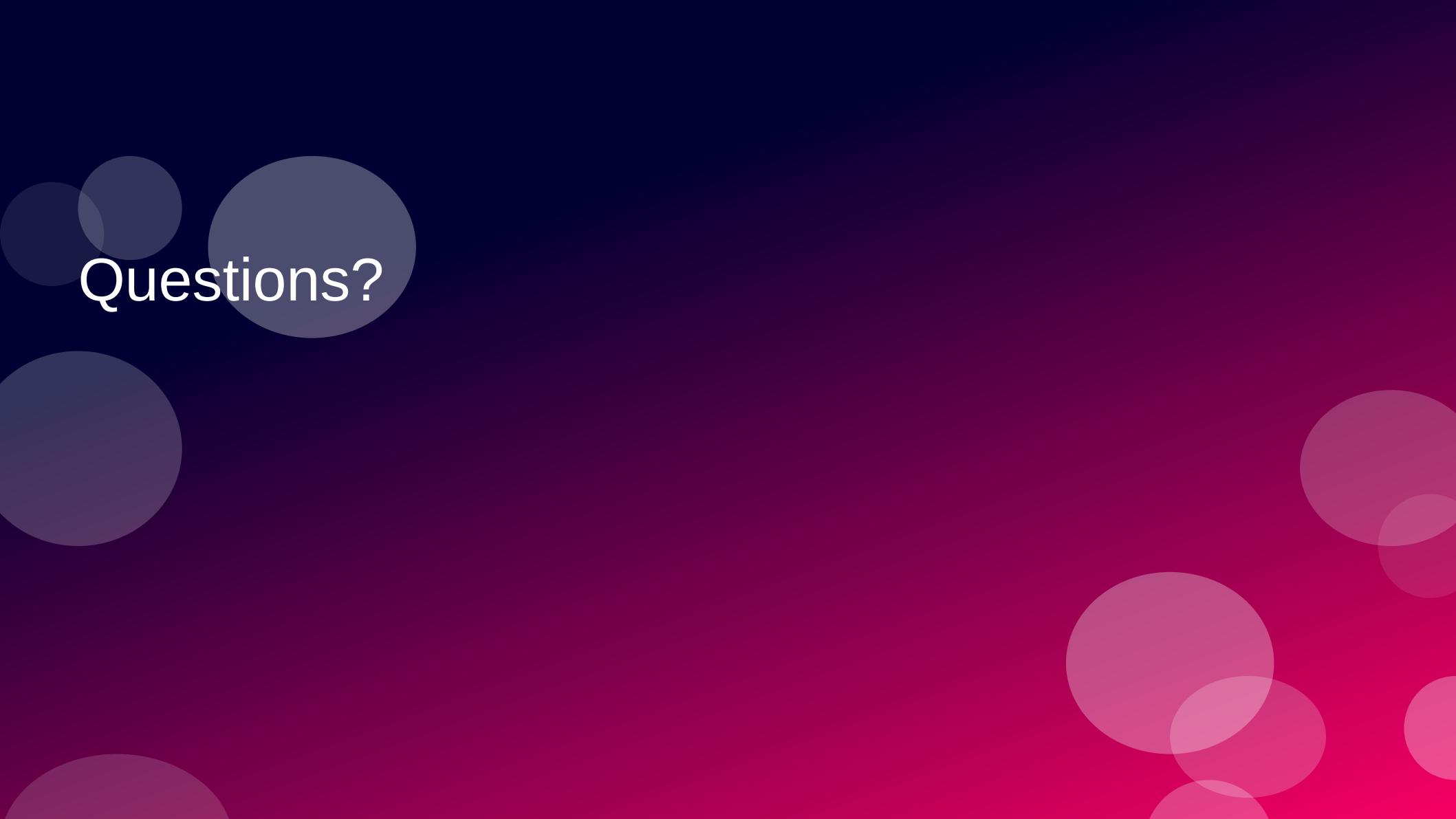
Energy



Conclusions



- Task-specialized morphologies can be generated using RL
- Generated morphologies show sufficient accuracy and greatly reduced energy consumption
- Fragile planner biases towards simple morphologies
- Would co-optimization of control and morphology remove that bias?
- Other learning techniques show promise.



Questions?

References

- [1] J. P. Madiedo, A. Chandrasekaran, and F. Salvador, "Capturing the benefits of worker specialization: Effects of managerial and organizational task experience," *Production and Operations Management*, vol. 29, no. 4, pp. 973–994, 2020.
- [2] R. V. Patel, "Opportunities and Challenges for Adoption of Robotics in Packaging Industry: A Review," *IOSR Journal of Engineering*, vol. 14, no. 1, pp. 32–38, Jan. 2024. Available: https://www.iosrjen.org/Papers/vol14_issue1/E1401013238.pdf
- [3] S. Haddadin et al., "The Franka Emika Robot: A Reference Platform for Robotics Research and Education," *IEEE Robotics & Automation Magazine*, vol. 29, no. 2, pp. 46–64, 2022. (Note: You can also append doi: 10.1109/MRA.2021.3138382 to the end if you prefer).
- [4] Standard Bots, "FANUC robot prices: Cost, models, and buying insights in 2025," Standard Bots, 2025. [Online]. Available: <https://standardbots.com/blog/fanuc-robot-price>
- [5] B. He, S. Wang, and Y. Liu, "Underactuated Robotics: A Review," *International Journal of Advanced Robotic Systems*, vol. 16, no. 4, pp. 1–14, 2019. (Optional: doi: 10.1177/1729881419862164)
- [6] K. S. Luck, H. B. Amor, R. Calandra, and J. Peters, "Data-efficient Co-Adaptation of Morphology and Behaviour with Deep Reinforcement Learning," in *Conference on Robot Learning (CoRL)*, 2019.
- [7] J. Bhatia et al., "Reinforcement Learning for Freeform Robot Design," 2023, arXiv:2310.05670.
- [8] Z. Ding, H. Tang, H. Wan, C. Zhang, and R. Sun, "A Modular Robotic Arm Configuration Design Method Based on Double DQN with Prioritized Experience Replay," *Symmetry*, vol. 16, no. 6, p. 714, 2024. [Online]. Available: <https://www.mdpi.com/2073-8994/16/6/714>
- [9] S. R. Buss, "Introduction to Inverse Kinematics with Jacobian Transpose, Pseudoinverse and Damped Least Squares methods," Dept. Math., Univ. California, San Diego, San Diego, CA, USA, unpubl. manuscript, 2004. Available: <http://www.math.ucsd.edu/~sbuss/ResearchWeb/ikmethods/iksurvey.pdf>
- [10] J. J. Kuffner and S. M. LaValle, "RRT-connect: An efficient approach to single-query path planning," in *Proceedings 2000 ICRA. Millennium Conference. IEEE International Conference on Robotics and Automation*, vol. 2, pp. 995–1001, 2000.
- [43] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal Policy Optimization Algorithms," 2017, arXiv:1707.06347.
- [42] J. Schulman, P. Moritz, S. Levine, M. I. Jordan, and P. Abbeel, "High-Dimensional Continuous Control Using Generalized Advantage Estimation," 2015, arXiv:1506.02438.
- [13] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor," in *Proceedings of the 35th International Conference on Machine Learning*, 2018.
- [14] N. Hansen and A. Ostermeier, "Completely Derandomized Self-Adaptation in Evolution Strategies," *Evolutionary Computation*, vol. 9, no. 2, pp. 159–195, 2001.